

Week #1: Basics

A. Military Enrollment Check

1 second, 256 megabytes

You want to enroll for military service. According to the requirements used in this problem:

- your height must be at least 160 cm;
- your weight must be at most 120 kg.

Given your height h and weight w , determine if you are accepted.

Input

The only line contains two integers h and w ($1 \leq h, w \leq 300$).

Output

Print `Accepted` if $h \geq 160$ and $w \leq 120$; otherwise, print `Rejected`.

input
173 65
output
Accepted

input
150 53
output
Rejected

input
160 120
output
Accepted

B. Bank Account Eligibility Counter

1 second, 256 megabytes

A bank allows opening a new account only if the customer's age is at least 21.

You are given the ages of n people. Count how many are eligible to open an account.

Input

The first line contains an integer n ($1 \leq n \leq 1000$).

The second line contains n integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq 120$) — the ages.

Output

Print one integer, the number of eligible people.

input
5 18 21 30 20 21
output
3

C. Factorial

1 second, 256 megabytes

You are given a number N . Print the factorial of N .

Input

The first line contains a number t ($1 \leq t \leq 20$), the number of test cases.

Next t lines will contain a number N ($0 \leq N \leq 20$).

Output

For each test case, print a single line containing the factorial of N .

input
3 1 5 3
output
1 120 6

D. Login Attempts

1 second, 256 megabytes

A system has a correct password (exactly 8 characters). A user can try to enter the password up to k times.

You will be given the correct password and then k attempted passwords. If any attempt matches the correct password exactly, print `Success` immediately. Otherwise, after all attempts, print `Locked`.

Input

The first line contains a string p — the correct password (its length is 8).

The second line contains an integer k ($1 \leq k \leq 20$).

Each of the next k lines contains a string s_i — an attempted password (its length is 8).

Output

Print `Success` if any attempt is equal to p ; otherwise, print `Locked`.

input
AbcD1234 3 aaaa1111 AbcD1234 zzzz9999
output
Success

input
AbcD1234 2 AbcD1235 AbcD1236
output
Locked

E. Shop Checkout

1 second, 256 megabytes

You are buying n items with prices $p_1, p_2, \dots, p_n$. Let S be the total sum of prices.

If $S \geq T$, you get a discount of $d\%$, and you pay the discounted total. Otherwise, you pay S .

Print the final bill. Your answer will be accepted if it has an absolute or relative error of at most 10^{-6} .

Input

The first line contains an integer n ($1 \leq n \leq 1000$).

The second line contains n numbers $p_1, p_2, \dots, p_n$ ($0 \leq p_i \leq 10^4$).

The third line contains two numbers T and d ($0 \leq T \leq 10^7$, $0 \leq d \leq 100$).

Output

Print one real number — the final bill.

Your answer will be considered correct if its absolute or relative error does not exceed 10^{-6} .

input
3 10 20 30 50 10
output
54.0000000000

input
2 10 20 40 10
output
30.0000000000

input
5 7 4 1 10 13 10 73
output
9.4500000000

F. Array Shift Check

1 second, 256 megabytes

You are given an array of n integers. Perform a right circular shift by k positions and print the resulting array.

A right circular shift by 1 moves the last element to the front. For example, shifting $[1, 2, 3, 4]$ by 1 becomes $[4, 1, 2, 3]$.

Input

The first line contains two integers n and k ($1 \leq n \leq 1000$, $0 \leq k \leq 10^9$).

The second line contains n integers $a_1, a_2, \dots, a_n$ ($-10^9 \leq a_i \leq 10^9$).

Output

Print n integers — the array after the right circular shift by k .

input
4 1 1 2 3 4
output
4 1 2 3

input
5 7 10 20 30 40 50
output
40 50 10 20 30

G. Function-Based Calculator

1 second, 256 megabytes

You are given two integers a and b and an operator character op .

Problems - Codeforces

Requirement: In your solution, implement four functions: `add(a,b)`, `sub(a,b)`, `mul(a,b)`, and `divi(a,b)`. Then use them to answer the query.

Compute:

- $a + b$ if op is `+`
- $a - b$ if op is `-`
- $a \times b$ if op is `*`
- $a \div b$ if op is `/`

If op is `/` and $b = 0$, print `Error`. Otherwise, print the integer result. For division, use integer division (as in C++).

Input

The only line contains two integers a and b and a character op ($-10^9 \leq a, b \leq 10^9$, $op \in \{+, -, *, /\}$).

Output

If division by zero happens, print `Error`. Otherwise, print the result.

input
7 5 +
output
12

input
3 4 *
output
12

input
5 2 -
output
3

input
10 3 /
output
3

input
10 0 /
output
Error

_Bounce_1. Recover Three Numbers

1 second, 256 megabytes

You are given three integers:

$A = x + y$

$B = y + z$

$C = x + z$

where x, y , and z are integers.

Your task is to recover and print the values of x, y , and z .

It is guaranteed that a valid integer solution exists.

Input

The only line contains three integers A, B , and C ($-10^9 \leq A, B, C \leq 10^9$).

Output

Print three integers x, y , and z .

input
3 5 4
output
1 2 3

_Bounce_2. Clean Title Case Line

1 second, 256 megabytes

You are given one line of text.

Split the line into words (words are separated by one or more spaces). For each word:

- make its first character uppercase,
- make all other characters lowercase.

Finally, concatenate all processed words together (remove all spaces) and print the result.

Input

The input contains one line s ($1 \leq |s| \leq 10^5$).

It is guaranteed that s contains only spaces and Latin letters (uppercase or lowercase), and it contains at least one letter.

Output

Print the cleaned line.

input
heLLo woRLD iti
output
HelloWorldIti

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